AI For Insightful Data Analysis

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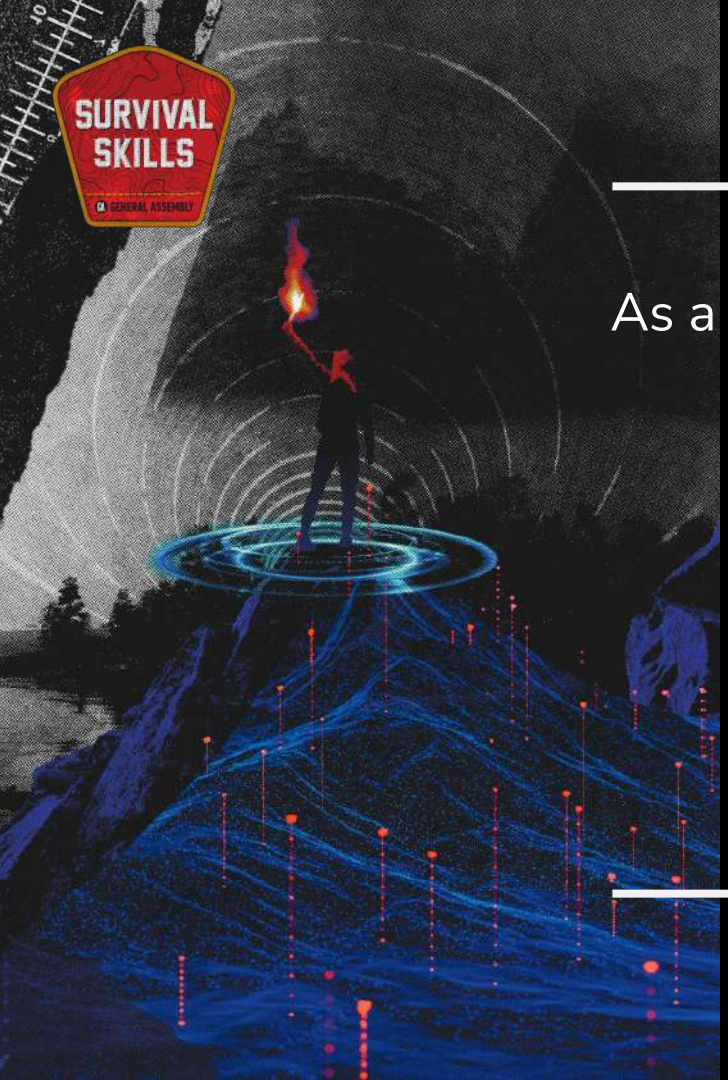


As a thank you for joining us today, we're offering:

$25 off a workshop with code Survive26

+ No application fee for new AI courses

Enroll by May 15, 2026

Terms & conditions apply. Discount amount varies based on region and course selected.
Mention Survival Skills when you speak to Admissions to redeem for a course.

Before we get started...

Some quick reminders

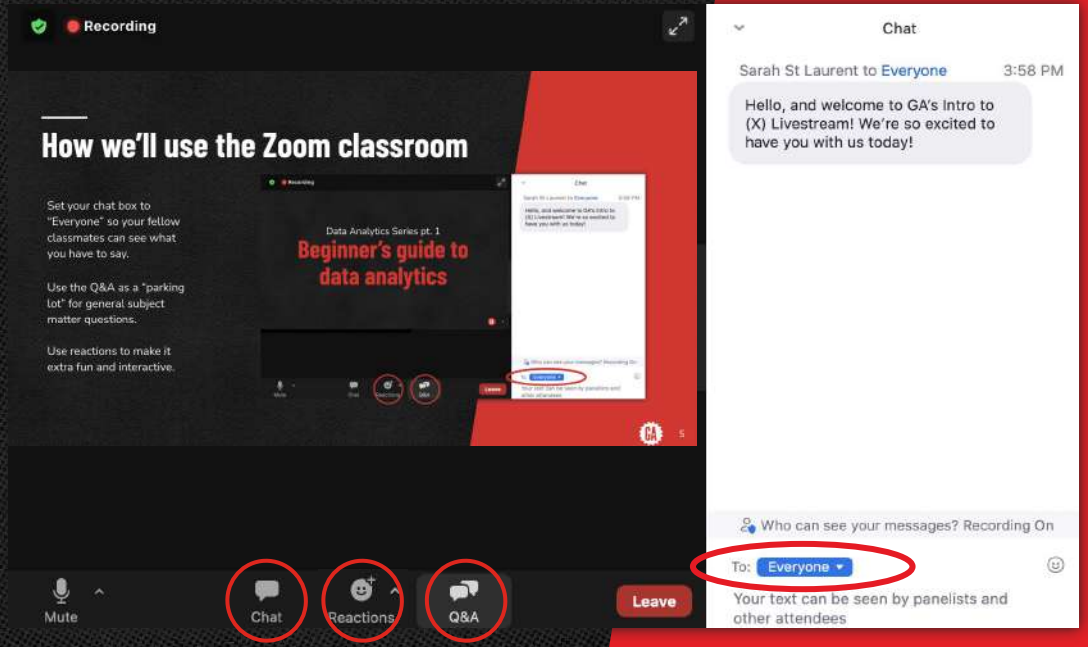
- This is a beginner-level class.
- There will be time for live Q&A at the end.
- YES—we're recording. You'll get the recording from the class via email in 24 hours.

How we'll use the Zoom classroom

Set your chat box to "Everyone" so your fellow classmates can see what you have to say.

Use the Q&A as a "parking lot" for general subject matter questions.

Use reactions to make it extra fun and interactive.



What you'll learn today

- Explore and analyze data with Generative AI
- Create compelling data visualizations using Generative AI
- Generate actionable insights with the help of Generative AI
- Revisit and clarify topics as needed

Now that we got that out of the way, let's dive in.

Using Gen AI for Data Analytics

What is Generative AI?

✨Generative AI

A form of Artificial Intelligence that uses generative models (like GPT-5) to produce text, images, videos or other forms of data.

📚Data Analytics with Gen AI:

Gen AI models are capable of writing code to analyze data sets, perform forecasting, producing charts and graphs, and also provide insights based on the data and analysis.

We often think of these Gen AI models as 'junior data analysts' who assists in the initial data exploration and cleaning work required within all data analytics workflows.

Core Benefits of Gen AI for Data Analysis

1

Increased efficiency and reduced human error.

2

Ability to uncover hidden patterns and trends which may be missed by humans.

3

Provide insights based on vast and diverse trained data.

4

Agentic AI's potential to perform tasks based on data-driven insights.

Use Cases of Generative AI for Data Analytics

1

Using Gen AI in Financial Services to fight fraud

Strengthening financial services with AI fraud detection

How financial institutions are using GenAI and real-time detection to stop fraud before it happens

By **Kelly Manrique**

29 April 2025

https://www.elastic.co/blog/financial-services-ai-fraud-detection

Use Cases of Generative AI for Data Analytics

2

Using Gen AI in Financial Services to fight fraud

DHL Supply Chain implements Generative AI to enhance data management, customer support and proposal accuracy

10/24/2024, 10:00 AM CEST

This strategic initiative marks a pivotal step in DHL Supply Chain's wider efforts to harness the potential of AI and to improve customer experience.

https://group.dhl.com/en/media-relations/press-releases/2024/dhl-supply-chain-implements-generative-ai.html

Use Cases of Generative AI for Data Analytics

3

Healthcare - Writing Hospitalisation Discharge Summaries

Integration of customised LLM for discharge summary generation in real-world clinical settings: a pilot study on RUSSELL GPT

Chun En Chua,[a] Ngoh Lee Ying Clara,[a,b,c,+] Mohammad Shaheryar Furqan,[b,c] James Lee Wai Kit,[b,c,d] Andrew Makmur,[b,c,e] Yih Chung Tham,[f] Amelia Santosa,[a] and Kee Yuan Ngiam[b,c,d]

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[c]Division of Biomedical Informatics, Yong Loo Lin School of Medicine, National University of Singapore, Singapore
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[f]Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore

https://www.thelancet.com/pdfs/journals/lanwpc/PIIS2666-6065(24)00205-0.pdf

Exploring Data

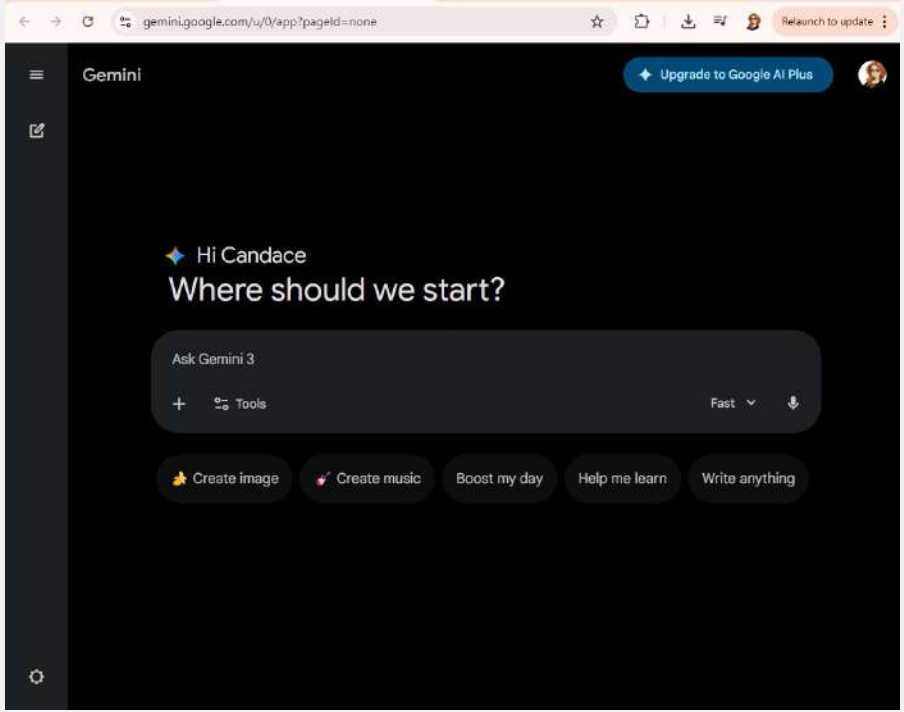
Introducing Gemini - Your AI Data Analyst

We will be using Gemini for our exercises!

- Uses natural language to analyze data
- Writes and explains analysis code for you
- Helps explore, visualize, and generate insights
- Requires human judgment and validation

Sign up for Gemini

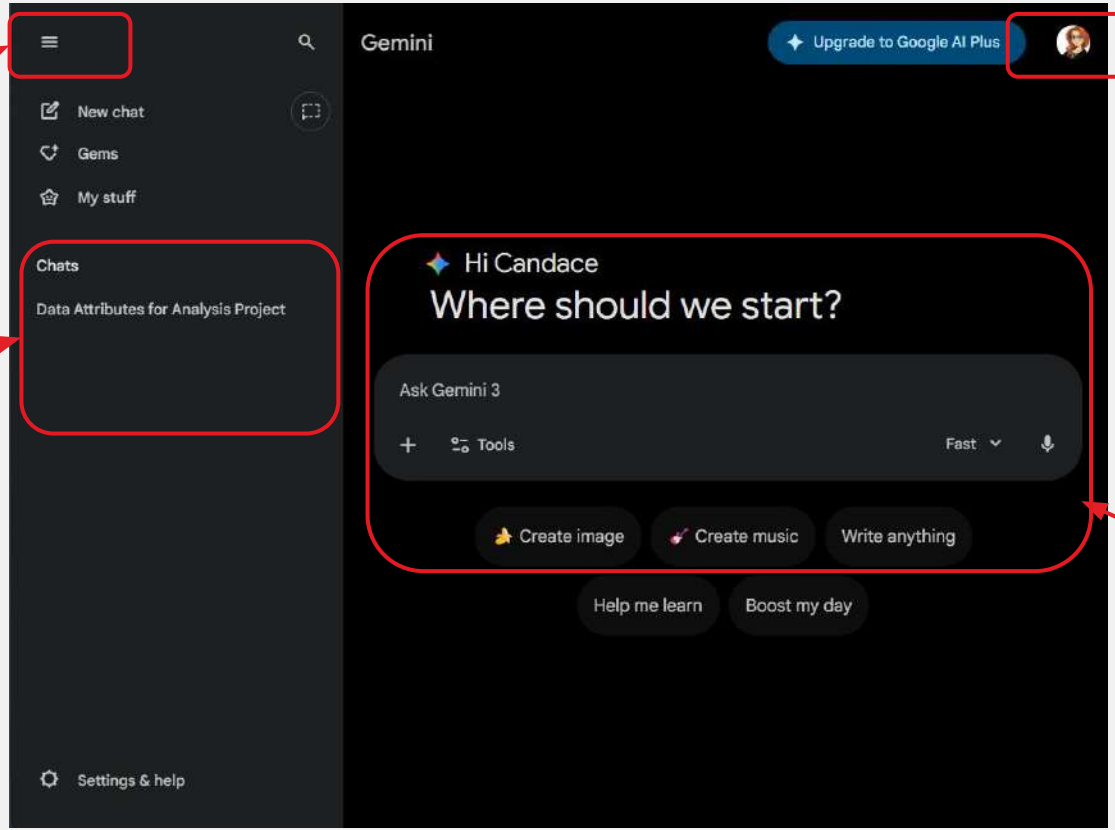
- Go to https://gemini.google.com/
- Sign in using your Google Account for free access



Gemini - Welcome Screen

Click on this hide the menu bar.

Conversation History



Log out of Google Account by clicking your avatar

We will focus mainly on this section as we go through our first few exercises.

Dataset - Absenteeism At Work

We will be working with the "Absenteeism At Work" dataset today.

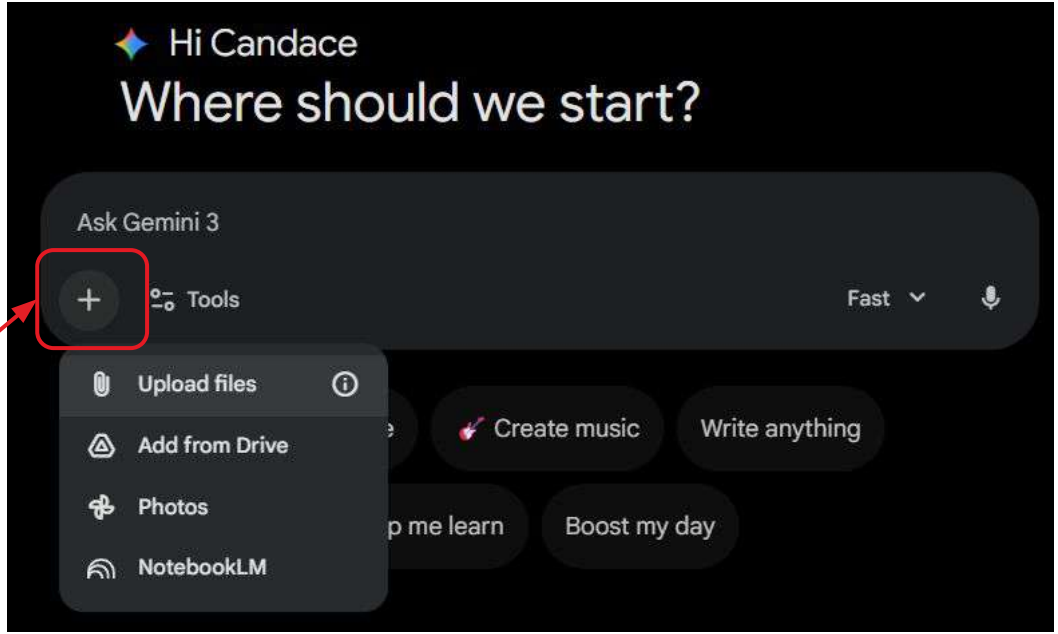
- Dataset is shared on Zoom. The original data can also be found here.
- The data is used in academic research at the Universidade Nove de Julho.
- It has 740 rows of data, and has 19 features. These features include:
  - Reason for absence
  - Month of absence
  - Day of week
  - Age
  - Education
  - Weight
  - Height
  - Body Mass Index
- For the purpose of this workshop, some cells have data removed to demonstrate 'missing data'.

Activities

Click on the **plus sign** to bring up various options.

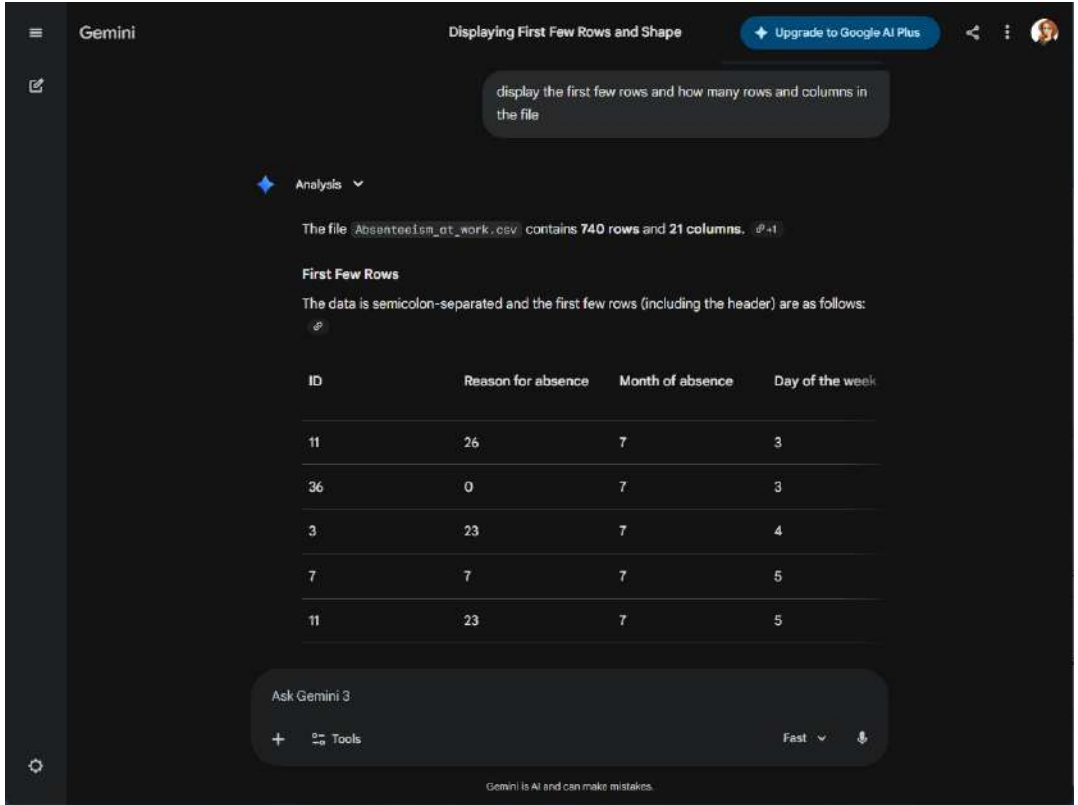
Click on **Add files.**
Find and **Upload Files** **‘Absenteeism_at_work.csv’.**

Add Files



Activities

Type **“display the first few rows** and how many **rows and columns in the file**” in the prompt and **click the arrow button.**



Gemini - Loading the file and showing the code used

To view the code used by Gemini to see the first few rows, ask: “Show the code used to generate the first few rows.”

Gemini

Displaying First Few Rows and Shape

Upgrade to Google AI Plus

Show the code used to generate the first few rows.

To generate the first few rows and get the dimensions of the file, you can use the **Python** `pandas` **library.**

Since the uploaded file `Absenteeism_at_work.csv` uses a semicolon (`;`) as a delimiter instead of a standard comma, you must specify the `sep` parameter.

Here is the code to achieve this:

Python

```python
import pandas as pd

# Load the dataset with the correct delimiter
df = pd.read_csv('Absenteeism_at_work.csv', sep=';')

# Display the number of rows and columns
rows, columns = df.shape
print(f"Total Rows: {rows}")
print(f"Total Columns: {columns}")

# Display the first five rows
print("\nFirst few rows of the dataset:")
print(df.head())
```

Explanation of the Code:

- `pd.read_csv(..., sep=';')`: This reads the CSV file into a DataFrame. The `sep=';'` is

Ask Gemini 3

Tools

Fast

Gemini is AI and can make mistakes.

GUIDED PRACTICE: Summary Statistics

Activities

Type “**Acting as a data analyst, show summary statistics**” in the **prompt** and click the arrow button. Scroll down to see more information.

Acting as a data analyst, show summary statistics

Analysis

As a data analyst, I have computed the summary statistics for the numerical attributes in the `Absenteeism_at_work.csv` dataset. These statistics provide a snapshot of the central tendency, dispersion, and range of the data.

Summary Statistics Table

Attribute	Mean	Std Dev	Min
Transportation expense	221.33	66.95	118
Distance from Residence to Work	29.63	14.84	5
Service time	12.55	4.38	1
Age	36.45	6.48	27
Work load Average/day	271.49	39.05	205.917

Export to Sheets

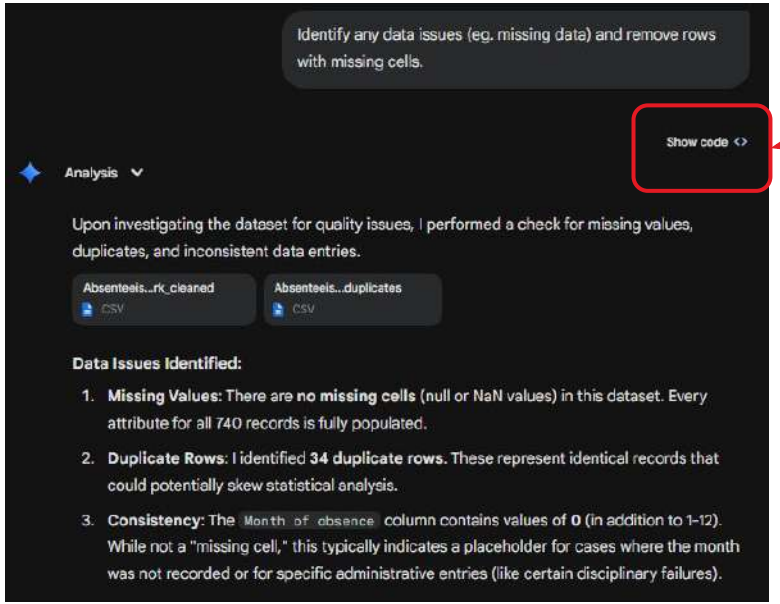
Key Analyst Insights:

- **Absenteeism Variance:** The "Absenteeism time in hours" shows a high standard deviation (13.33) relative to its mean (6.92), indicating significant variability in how long employees are absent. While the average is around 7 hours, some instances reach up to 120 hours.
- **Employee Demographics:** The average employee is approximately 36 years old with a BMI of 26.68 (falling within the "overweight" category). +1
- **Commute Factors:** There is a wide range in transportation expenses (118 to 388 units) and distance to work (5 to 52 km), which may be potential factors to explore in relation to absenteeism frequency. +1
- **Target Performance:** The "Hit target" percentage remains high across the board, with a mean of 94.59% and a minimum of 81%, suggesting that absenteeism may not have a catastrophic impact on immediate goal achievement.

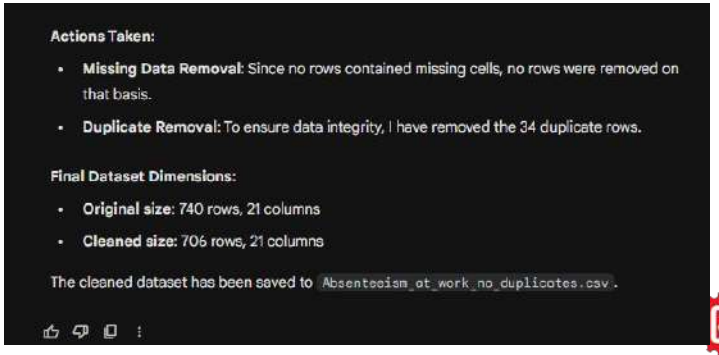
Sources

Activities

Type **“Identify any data issues (eg. missing data)** and **remove rows with missing cells.**” in the prompt and click the arrow button. Scroll to see more.



Another option to see the code generated for the analysis



GUIDED PRACTICE: 5 - Explore by summarising the feature columns in the cleaned dataset.

Activities

Type "**Provide a summary of the different features in this cleaned dataset**." in the prompt and click the blue up arrow button.

Provide a summary of the different features in this cleaned dataset

Show code

Analysis

The cleaned dataset contains **21 features** that can be categorized into four logical groups. Following the removal of duplicates, we have **706 records** of employee data.

1. Target Variable

- **Absenteeism time in hours:** The duration of absence. It ranges from **0 to 120 hours**, with 19 unique duration values recorded.

2. Temporal & Administrative Factors

These features provide context on *when* and *why* the absence occurred.

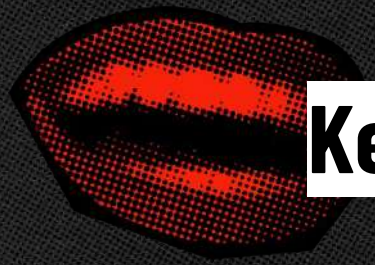
- **Reason for absence:** Categorized into **28 types**. Codes **1–21** correspond to the International Code of Diseases (ICD), while 22–28 cover reasons like dental visits or physiotherapy.
- **Month of absence:** Values **0 to 12** (0 acts as a placeholder/unknown).
- **Day of the week:** Values **2 to 6** (representing Monday through Friday).
- **Seasons:** 4 categories (Summer, Autumn, Winter, Spring).
- **Disciplinary failure:** A binary indicator **(0 or 1)** of whether the employee has a history of disciplinary issues.

- **Weight & Height:** Physical measurements used to calculate BMI.
- **Social Drinker/Smoker:** Binary flags **(0 or 1)** indicating lifestyle habits.

Feature Summary Table (Cleaned Data)

Category	Feature Count	Key Metrics
Total Features	21	1 Target, 20 Predictors
Binary Flags	3	Disciplinary failure, Social drinker, Social smoker
Numerical/Continuous	10	Age, Service time, Distance, BMI, Work load, etc.
Categorical/Nominal	8	ID, Reason, Month, Day, Season, Education, etc.

Export to Sheets



Key takeaways

Using plain English to instruct the Gen AI.

No code!

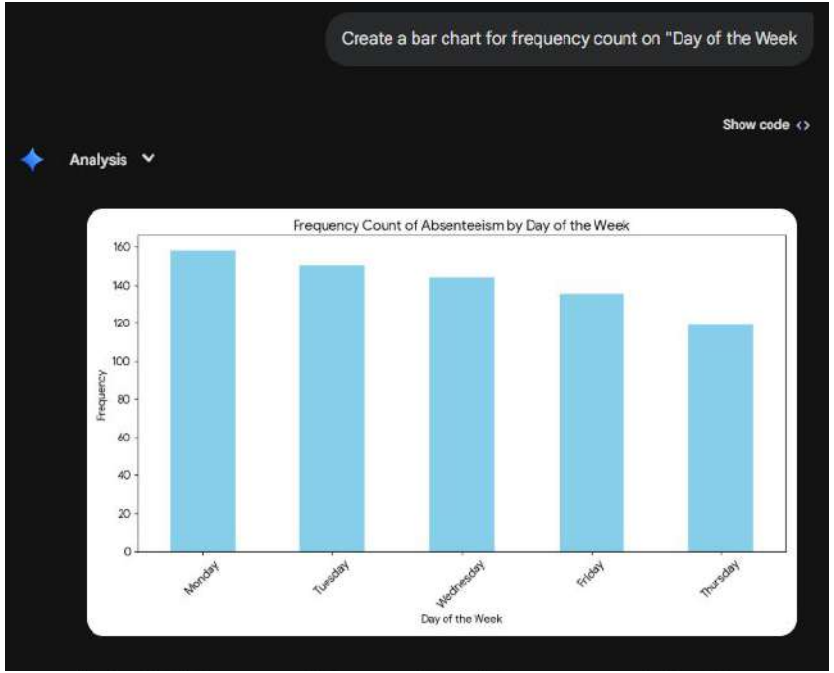
Gen AI provides useful suggestions for next steps.

Visualizing Data

GUIDED PRACTICE: 6 - Creating bar charts to find which day of the week has the most absentees.

Activities

Type “**Create a bar chart for frequency count on "Day of the Week".**” in the prompt and click the arrow button. Scroll to see more.



The bar chart below displays the frequency of absenteeism records for each day of the week.

Frequency of Absenteeism by Day

According to the dataset, Monday and Tuesday show the highest frequency of recorded absences, while Thursday has the lowest.

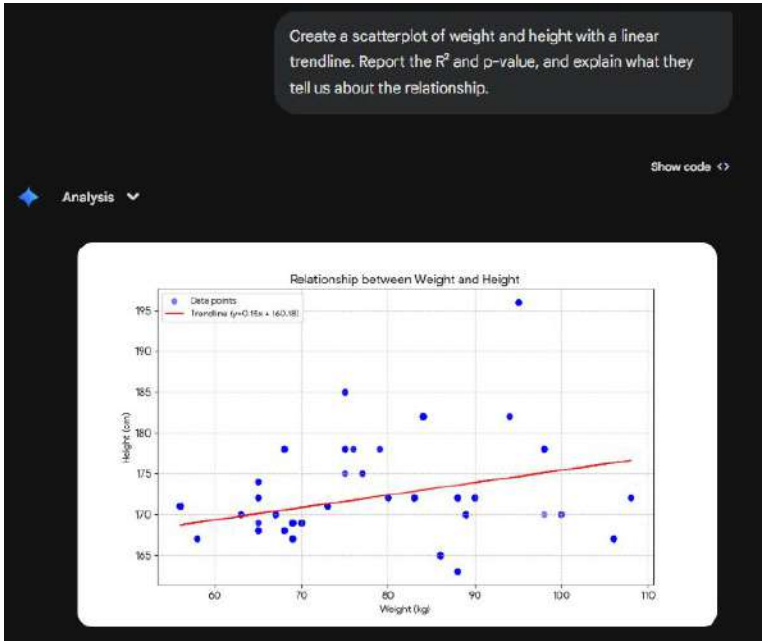
Day of the Week	Number of Records
Monday	158
Tuesday	150
Wednesday	144
Friday	135
Thursday	119

Export to Sheets

(Note: The days are mapped based on the dataset documentation where 2=Monday, 3=Tuesday, 4=Wednesday, 5=Thursday, and 6=Friday.)

Activities

Type "**Create a scatterplot of weight and height with a linear trendline. Report the R² and p-value, and explain what they tell us about the relationship.**" in the prompt and click the arrow button.



The scatterplot below illustrates the relationship between the weight and height of the employees in our dataset.

Statistical Summary

- R^2 **(Coefficient of Determination):** 0.1010
- **p-value:** 4.9564×10^{-18}

Interpretation of Results

1. What the R^2 tells us:
The R^2 value of 0.1010 indicates that approximately **10.1%** of the variance in height can be explained by weight. In the context of data analysis, this is considered a **weak linear relationship**. While height and weight are naturally related, many other factors (such as genetics and age) influence height, and the high variation in body types among employees means weight is not a strong predictor of height in this specific group.

2. What the p-value tells us:
The p-value is extremely small (< 0.05), which means the relationship is **statistically significant**. This indicates that the positive correlation we see is very unlikely to have occurred by random chance. Even though the relationship is weak (as shown by the R^2), we can be highly confident that a trend exists: generally, as weight increases, height also tends to increase.

3. The Trendline:
The trendline equation is approximately $Height = 0.15 \times Weight + 160.18$. This suggests

Key takeaways

Gen AI can construct various charts and graphs

No code!

Can download charts, and written summary of chart findings.

Creating Insights

Activities

Type "**Provide 5 useful insights based on the data**." in the prompt and click the arrow button.

Provide 5 useful insights based on the data

Show code

Analysis

As a data analyst, I have synthesized the following 5 key insights from the cleaned dataset. These findings highlight the primary drivers of absenteeism and offer actionable focus areas for organizational wellness.

1. High-Impact vs. High-Frequency Reasons

While "Medical Consultations" (Reason 23) and "Dental Consultations" (Reason 28) are the **most frequent** causes of absence, they result in very short durations (avg. ~3 hours). In contrast, **Diseases of the musculoskeletal system** (Reason 13) and **Injuries/Poisoning** (Reason 19) are the primary drivers of **total lost productivity**, accounting for the highest cumulative hours.

- **Actionable Insight:** The company might benefit from ergonomic assessments or physical therapy benefits to address musculoskeletal issues.

2. The Lifestyle Factor: Social Drinking

The data reveals a notable difference in absenteeism based on lifestyle choices. **Social drinkers** average approximately **7.95 hours** of absence per instance, whereas non-drinkers average only **6.09 hours**. This suggests that lifestyle-related health or recovery may play a role in work attendance patterns.

3. Seasonal Sensitivity (The "Winter Peak")

Absenteeism is not spread evenly across the year. **Winter (Season 3)** shows the highest average absence duration at **8.36 hours** per instance, compared to Autumn and Spring (~6.4 hours).

- **Actionable Insight:** This peak suggests a need for increased health awareness or flu vaccination programs leading into the winter months.

4. Education and Work Responsibility

There is an inverse relationship between education level and absenteeism duration. Employees with **Master's or PhD levels (Education 4)** have the lowest average absence (5.25 hours), while those with only a **High School education (Education 1)** average nearly 40% more time away (7.43 hours). This may correlate with job roles, levels of responsibility, or health literacy.

5. The "Distance Paradox"

Surprisingly, the correlation between "Distance from Residence to Work" and absenteeism time is **slightly negative (-0.08)**. This suggests that employees living further away do not necessarily miss more work hours than those living nearby. Instead, factors like **Age** and **Service Time** show a stronger (though still modest) positive relationship with absenteeism, indicating that tenure and life stage may be more relevant predictors than commute length.

Activities

Type "**Provide 5 recommended actions to address absenteeism based on the data**." in the prompt and click the up arrow button.

Provide 5 recommended actions to address absenteeism based on the data

As a data analyst, I recommend the following five strategic actions to reduce the impact of absenteeism at the company, based on the specific drivers identified in the dataset:

1. Implement Ergonomic and Workplace Safety Programs

The data shows that while medical consultations are frequent, **musculoskeletal diseases (Reason 13)** and **injuries (Reason 19)** are responsible for the highest volume of total hours lost.

- **Action:** Conduct ergonomic assessments for all workstations and provide specialized training on safe lifting or posture. Investing in better office furniture or protective equipment could significantly reduce these long-duration absences.

2. Introduce "On-Site" or Telehealth Medical Services

"Medical consultations" and "Dental consultations" are the most frequent reasons for absence, but they typically last only a few hours. This suggests employees are leaving work for routine appointments.

- **Action:** Partner with a telehealth provider or host monthly on-site dental/health check-up vans. Reducing the travel time required for these routine visits can convert half-day absences into one-hour breaks, recovering significant productive time.

3. Launch Seasonal Wellness Campaigns (Focus on Winter)

The analysis identified **Winter (Season 3)** as the period with the highest average absence duration per instance.

- **Action:** Proactively launch "Winter Wellness" campaigns starting in late Autumn. This should include on-site flu vaccinations, vitamin D awareness, and perhaps temporary flexible "work-from-home" options during peak illness weeks to prevent the spread of contagious viruses within the office.

4. Promote Lifestyle Wellness & Responsible Habits

The dataset highlights that **social drinkers** have higher average absenteeism hours (~8 hours) compared to non-drinkers (~6 hours).

- **Action:** Incorporate "Healthy Lifestyle" modules into the corporate wellness program. This doesn't require a ban on social activities but can involve education on the impact of alcohol on sleep and recovery, or offering incentives for participating in fitness challenges

5. Review Support Systems for Lower-Education Job Roles

There is a clear trend where employees with lower education levels (typically in more physical or repetitive roles) experience higher absenteeism.

- **Action:** Investigate the specific physical or mental strain associated with these roles. Providing more frequent short breaks, "mental health days," or career development support for these specific groups could improve job satisfaction and physical longevity, thereby reducing the need for extended sick leave.

GUIDED PRACTICE: 10 - Write a concise report to on absenteeism at the workplace based on this dataset.

Activities

Type “**You are a data journalist. Write a concise journalistic report on absenteeism at the workplace based on this dataset. Format like an online article.**” in the prompt and click the up arrow button.

You are a data journalist. Write a concise journalistic report on absenteeism at the workplace based on this dataset. Format like an online article.

SPECIAL REPORT: The Hidden Cost of the Work Week

By Data Insights Desk *Estimated reading time: 3 minutes*

New analysis of courier industry data has revealed the complex anatomy of employee absenteeism, suggesting that the "why" and "when" of missing work are far more nuanced than simple sick days. By examining over 700 records from a prominent courier firm, data journalists have identified the key drivers keeping employees away from their desks.

The Monday Blues are Real

The data confirms a long-standing office suspicion: absenteeism peaks at the start of the week. **Monday** leads as the most frequent day for absences (158 instances), followed closely by Tuesday. As the week progresses, attendance stabilizes, reaching its most reliable point on **Thursday** before a slight uptick on Friday.

Chronic Pain vs. Routine Care

While the most frequent reasons for absence are routine medical and dental consultations—short-duration visits that often last only a few hours—the real "productivity killers" are musculoskeletal issues.

Records show that diseases of the musculoskeletal system and connective tissue (ICD Category XIII) account for the highest cumulative volume of lost hours. While an employee might nip out for an hour for a dental check-up, a back injury or chronic joint issue can sideline

Key takeaways

Ask Gen AI to create insights based on the data.

Gen AI can provide recommendation actions based on the data and its own training data.

Let Gen AI help you write reports and articles with the right prompt!

Reflection

Incorporate Gen AI into Your Workflows

Exploratory Data Analysis	Great tool for initial exploration and summary of data
Identify Data Issues	Identify issues with data (eg. missing cells)
Summarise Data	Create summaries of dataset
Data Visualization	Generate charts and graphs!
Generate Insights	Generate data-driven Insights

Gotchas!

Private Data

Adhere to the appropriate data regulations regarding Personal Identifiable Information (PII)

Ethics

Consider the purpose and intent of performing your analysis. Make sure it does no harm to others.

Sensitive Data / Information

Some data within your organisation may be confidential. Use mock data in place of sensitive data.

Review Gen AI Output

Gen AI can make mistakes and hallucinate. Have processes in place to review Gen AI output!

Q & A

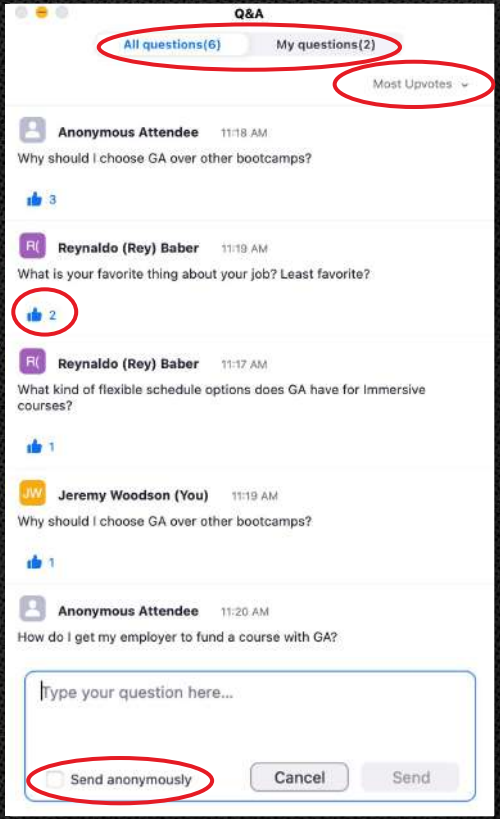
View all questions, or only questions you've asked.

Sort by Most Upvotes or Recently Asked

Already see your question posted? Avoid duplicates and **upvote it!**

Send anonymous questions if you're more comfortable. Otherwise, let us know who you are.

You're up.



Let's Wrap up!

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Let us know what you thought of today's event:

ga.co/skillsfeedback

Additional Resources

PRACTICE AGAIN	DIGGING DEEPER
Data Sources • UCI Irvine Machine Learning Repo • Kaggle **GPTs** • Google Gemini • OpenAI ChatGPT • Anthropic Claude	**Articles** • AI-Driven Data Analytics • How I Embraced AI for Data Analysis • Generative AI in Data Analytics

Thank You

GENERAL ASSEMBLY

Can’t wait to see you next time.